Planet Hunters

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HAT-P-7b: Abstract

- Extremely tilted orbit, either retrograde or nearly polar.
- Evidence for an additional planet or companion star.
- Possible explanations.

HAT-P-7 System: Model Parameters

HAT-P-7: F6V star, $M = 1.5M_{\odot}$, $R = 1.8R_{\odot}$.
HAT-P-7b: $M = 1.8M_{\mathrm{Jup}}$, $R = 1.2R_{\mathrm{Jup}}$.

Parameter	Value
Orbital period, P (d)	2.2047304 ± 0.0000024
Midtransit time (HJD)	$2454{,}731.67929 \pm 0.00043$
Transit duration (first to fourth contact) (hr)	4.006 ± 0.064
Transit ingress or egress duration (hr)	$0.474^{+0.061}_{-0.093}$
Planet-to-star radius ratio, $R_p/R_\star$	$0.0834^{+0.0012}_{-0.0021}$
Orbital inclination, i (deg)	$80.8^{+2.8}_{-1.2}$
Scaled semimajor axis, $a/R_\star$	$3.82^{+0.39}_{-0.16}$
Transit impact parameter	$0.618^{+0.039}_{-0.149}$
Velocity semiamplitude, K (m s^{-1})	211.8 ± 2.6
Upper limit on eccentricity (99.73% conf.)	0.039
$e \cos \omega$	-0.0019 ± 0.0077
$e \sin \omega$	0.0037 ± 0.0124
Velocity offset, Keck/HIRES (m s^{-1})	-51.2 ± 3.6
Velocity offset, Subaru/HDS (m s^{-1})	-4.8 ± 2.5
Constant radial acceleration $\dot{\gamma}$ (m s^{-1} yr^{-1})	21.5 ± 2.6
Projected stellar rotation rate, $v \sin i_\star$ (km s^{-1})	$4.9^{+1.2}_{-0.9}$
Projected spin–orbit angle, λ (deg)	182.5 ± 9.4

A typical Hot-Jupiter

HAT-P-7b: Evidence for a Third Body

- Adding a parameter γ representing an extra radial acceleration fits better.
- The RVs from 2009 are systematically redshifted by approximately 40 m/s compared to RVs from 2007.
- An order-of-magnitude relation:

$$\frac{M_c \sin i_c}{a_c^2} \sim \frac{\dot{\gamma}}{G} = (0.121 \pm 0.014) M_{\text{Jup}}\,\text{AU}^{-2}$$

where M_c is the companion mass, i_c its orbital inclination, a_c its orbital distance.

? Why do they think the additional companion produce a radial acceleration? Is this assumption only for simplicity?

HAT-P-7b: Evidence for a Third Body

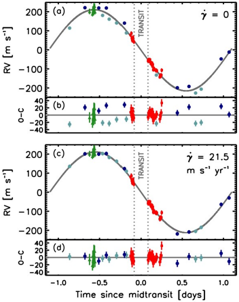


Figure 2. Phased radial velocity variation of HAT-P-7. (a) Assuming a single Keplerian orbit. (b) Residuals. (c) With an extra parameter $\dot{\gamma}$ representing a constant radial acceleration. (d) Residuals. The circles are HIRES data (light blue from 2007, dark blue from 2009), the green triangles are HDS data from 2009 June 17, and the red squares are HDS data from 2009 July 1.

Phased radial velocity variation of HAT-P-7

HAT-P-7b: Evidence for a Spin–orbit Misalignment

- The inversed Rossiter – McLaughlin effect which indicates that the orbital “north pole” and the stellar “north pole” point in nearly opposite directions on the sky.
- **Rossiter – McLaughlin effect:** A transiting planet in a well-aligned prograde orbit would first pass in front of the blueshifted (approaching) half of the star, causing an anomalous redshift of the observed starlight. Then, the planet would cross to the redshifted (receding) half of the star, causing an anomalous blueshift.

HAT-P-7b: Radial Velocities

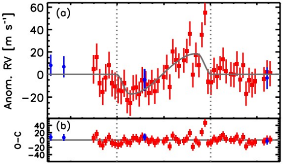


Figure: The anomalous RV

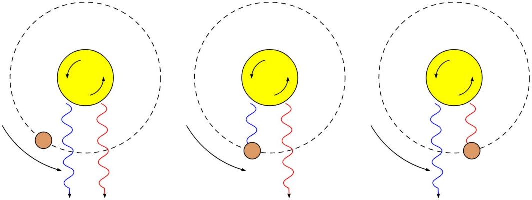


Figure: The Rossiter – McLaughlin effect

HAT-P-7b: Evidence for a Spin–orbit Misalignment

- ? Why do they need to do a photometry observation? Maybe we could look at Winn et al. 2009
- ? The difference between λ and ψ. Why they need λ?

HAT-P-7b: Discussion

- Close encounters between planets throw a planet inward, where its orbit is ultimately shrunk and circularized by tidal dissipation.
- Kozai effect(as $L = \sqrt{1 - e^2}\cos i$ is a constant, highly inclined orbits can become very eccentric): the gravitational force from a distant body on a highly inclined orbit strongly modulates an inner planet's orbital eccentricity and inclination.
- An inward-migrating outer planet that captures an inner planet into a mean motion resonance.

KIC 8462852: Abstract

- T. S. Boyajian et al. discovered that star KIC 8462852 has unusually and maybe irregularly big dips of its flux, sometimes over 20%.
- Fernando J. Ballesteros suggests that a large, ringed body whose transit produces the first dimming and a swarm of Trojan objects sharing its orbit that causes the second period of multiple dimmings.
- Peter Foukal explains it as convective stars can store the required fluxes.

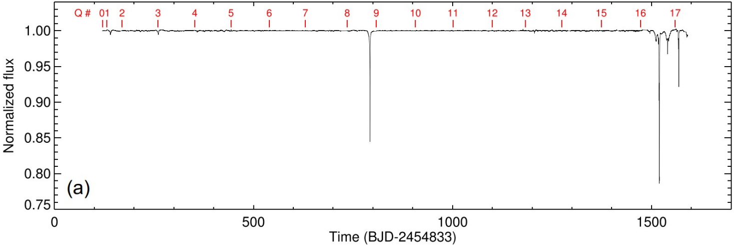
KIC 8462852: Properties

Estimates from spectroscopy:

$$M = 1.43M_\odot, \quad L = 4.68L_\odot, \quad R = 1.58R_\odot, \quad T_{\text{eff}} = 6750\,\text{K}$$

A main-sequence F3V star.

KIC 8462852: Where's the flux?

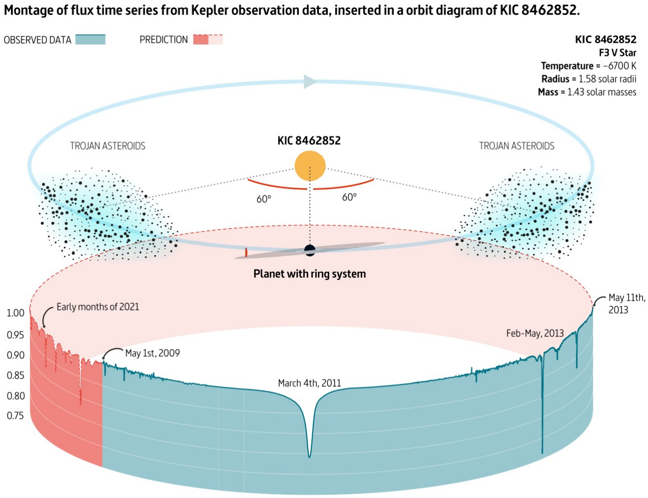


Dip no.	Name	Depth	BJD (−245 4833)	Cycles (from dip 5)	|Residual| (from integer)
1	(D140)	0.5 per cent	140.49	−13	0.52
2	(D260)	0.5 per cent	261.00	−11	0.01
3	(D360)	0.2 per cent	359.11	−9	0.04
4	(D425)	0.2 per cent	426.62	−7	0.44
5	(D800)	16 per cent	792.74	0	0.00
6	(D1200)	0.4 per cent	1205.96	8	0.54
7	(D1500)	0.3 per cent	1495.97	14	0.53
8	(D1520)	21 per cent	1519.60	15	0.02
9	(D1540)	3 per cent	1540.40	15	0.45
10	(D1570)	8 per cent	1568.49	16	0.03

KIC 8462852: Main observations

- Spectroscopy: gives the properties of the star.
- Imaging: found a companion star. (KIC 8462852 B is proved not a orbiting binary star, see Clemens et al. 2018 and Pearce et al. 2021.)
- Radial velocity: limits on a close companion, which shows that that there are no objects heavier than a super-Jupiter in close orbits with $P \lesssim 2\,\mathrm{d}$, and likely no heavier in mass than a brown dwarf for $P \lesssim 300\,\mathrm{d}$.

KIC 8462852: Planet and its Trojans?



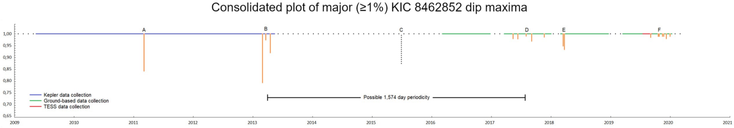
KIC 8462852: Convective Star?

- Thermal plugs
- The diverted heat is stored as a very small global heating and expansion of that envelope, with negligible influence on the photospheric temperature and luminosity.
- Magnetic activity

KIC 8462852: Other possibilities?

- Large comets: Bodman & Quillen 2016.
- Gas exoplanets and its gas moons: Martinez et al.

? Dyson Swarm?



A) 'D800' dip (Boyajian et al. 2016); B) 'D1500' dip cluster (Boyajian et al. 2016); C) possible missed reoccurrence of D800 dip (Sacco et al. 2017);
D) 'Elsie' dip cluster (Boyajian et al. 2018); E) ' Caral-Supe' and 'Evangeline' dips;
F) September-December 2019 dip cluster, possible reoccurrence of D800 dip (Sacco et al. 2017, Gary 2020)

Thank you for listening!